2. ACIDS, BASES AND SALTS SOLUTIONS

1. What is neutralization reaction? Give two examples.
Answer:
Neutralization reaction is a reaction in which an acid reacts with a base to form salt and water (usually with release of heat).
Examples:
1) $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$
2) $H_2SO_4(aq) + 2KOH(aq) \rightarrow K_2SO_4(aq) + 2H_2O(l)$

2. What is observed when carbon dioxide gas is passed through limewater (a) for a short duration? (b) for a long duration? Also write the chemical equations.
Answer:
(a) For a short duration: Limewater turns milky due to formation of insoluble calcium carbonate ($CaCO_3$).
$Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s)\downarrow + H_2O(l)$
(b) For a long duration (excess CO_2): The milkiness disappears because calcium carbonate reacts with excess CO_2 and water to form soluble calcium hydrogen carbonate.
$CaCO_3(s) + CO_2(g) + H_2O(l) \rightarrow Ca(HCO_3)_2(aq)$

3. While diluting an acid, why is it recommended that acid should be added to water and not water to the acid?
Answer:
Dilution of acid is highly exothermic (releases heat). If water is added to concentrated acid, the heat produced may cause the water to boil suddenly and splash acid out, leading to burns. Thus, always add acid slowly to water with continuous stirring so that heat is absorbed safely by the larger amount of water and splashing is avoided.

4. What is universal indicator? State the purpose for which this indicator is used.
Answer:
Universal indicator is a mixture of several indicators that shows different colours at different pH values.
Purpose: It is used to determine how strongly acidic or basic a solution is, i.e., to find the approximate pH of a solution by comparing the colour produced with a pH colour chart.

5. How does the enamel of the teeth undergo damage due to eating of chocolates and sweets? What should be done to prevent it?
Answer:
Bacteria present in the mouth break down sugar from chocolates and sweets to form acids. These acids lower the pH in the mouth below about 5.5, causing tooth enamel (mainly calcium phosphate) to dissolve, leading to tooth decay.
Prevention:
- Rinse mouth after eating sweets and brush teeth twice a day.
- Use basic toothpaste (contains mild bases) to neutralize acids.
- Limit frequent intake of sugary foods.

6. Differentiate between strong electrolytes and weak electrolytes. Mention two examples.
Answer:

Basis	Strong electrolyte	Weak electrolyte
Ionisation in water	Almost completely ionises (forms ions)	Partially ionises (forms fewer ions)
Electrical conductivity	Good conductor of electricity	Poor conductor of electricity

Examples:
- Strong electrolytes: $HCl(aq)$, $NaOH(aq)$, $NaCl(aq)$ (any two).
- Weak electrolytes: $CH_3COOH(aq)$, $NH_4OH(aq)$ (any two).

7. (a) In a tabular form write the colours of the following indicators in presence of acid and base: Litmus solution, phenolphthalein solution, methyl orange solution.
(b) Classify the following solutions A and B as acidic/basic, giving reason.
Solution A: $[H^+] < [OH^-]$; Solution B: $[H^+] > [OH^-]$
Answer:

Indicator	In Acid	In Base
Litmus	Red	Blue
Phenolphthalein	Colourless	Pink
Methyl orange	Red	Yellow

(b) Classification:

· Solution A is basic because [H+] < [OH−] (more hydroxide ions than hydrogen ions).

· Solution B is acidic because [H+] > [OH−] (more hydrogen ions than hydroxide ions).

8. Give the formula of bleaching powder and write two applications of it.

Answer:

Formula of bleaching powder: $CaOCl_2$ (calcium oxychloride)

Applications (any two):

· Used for bleaching cotton and linen in textile industry, and wood pulp in paper industry.

· Used for disinfecting drinking water (kills germs).

· Used as an oxidising agent in chemical industries (also acceptable).

9. How is Plaster of Paris chemically different from gypsum? How can they be interconverted? Write any two uses of Plaster of Paris.

Answer:

Chemical difference:

- Gypsum is calcium sulphate dihydrate:

$CaSO_4 \cdot {}_2H_2O$

- Plaster of Paris (POP) is calcium sulphate hemihydrate:

$CaSO_4 \cdot ({}_1/_2)H_2O$

Interconversion:

1. 1) Gypsum → POP (on heating at about 373 K):

$CaSO_4 \cdot 2H_2O$ —heat→ $CaSO_4 \cdot (1/2)H_2O + (3/2)H_2O$

2. 2) POP → Gypsum (on adding water, setting of POP):

$CaSO_4 \cdot (1/2)H_2O + (3/2)H_2O \rightarrow CaSO_4 \cdot 2H_2O$

Uses of POP (any two):

- Used for making casts for fractured bones.
- Used for making toys, decorative items, statues, and false ceilings.

10. What is chlor-alkali process? Write its chemical equation. Which gas is evolved at cathode and anode? Write any two uses of hydrochloric acid and bleach so formed as products.

Answer:

Chlor-alkali process is the electrolysis of aqueous sodium chloride (brine) to produce chlorine (chlor), sodium hydroxide (alkali) and hydrogen.

Overall chemical equation:

$2NaCl(aq) + 2H_2O(l)$ —electricity→ $2NaOH(aq) + Cl_2(g) + H_2(g)$

Gases evolved:

· At anode: chlorine gas (Cl_2).

· At cathode: hydrogen gas (H_2).

Uses (any two each):

· Hydrochloric acid (HCl): used for cleaning metals/pickling before galvanising; used in manufacture of chloride salts; used as laboratory reagent (any two).

· Bleach (chlorine/bleaching powder): bleaching cotton/linen and wood pulp; disinfecting drinking water; making bleaching powder and other chemicals (any two).